

1. Overview & specifications

The Frostline Inverter Split AC XYZ-492 (Frostline, air-conditioner) is covered by this service manual for authorised technicians holding the 'inverter-ac' skill. It documents diagnosis, fault codes, and board/mechanical-level repair.

Specifications — type: split inverter AC, capacityTon: 1.5, refrigerant: R32, powerW: 1480, starRating: 5.

Warranty: 60 months. Parts and labour on the inverter control board (PCB) and compressor. Consumables excluded.

Reference: <https://support.example-fieldflow.test/xyz-492/manual>

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2. Installation & commissioning

Mount the indoor unit level, flare and torque the refrigerant lines, pull a vacuum to 500 microns, then release the R32 charge and confirm subcooling.

Record the serial number and commissioning date against the work order. A failed self-test must be cleared before the visit is closed.

3. Fault codes

F3 — Inverter control board failure; compressor will not start. Replace PCB-492. Skill: inverter-ac.

E1 — High-voltage capacitor fault; discharge before service. Replace CAP-492. Skill: inverter-ac.
SAFETY: high-risk, follow discharge/isolation SOP.

F6 — Outdoor temperature sensor open circuit. Replace SENSOR-11. Skill: general-hvac.

U4 — Indoor/outdoor communication error; inspect harness first. No part required; inspect and reset. Skill: general-hvac.

P4 — Compressor over-current trip. Replace COMP-492. Skill: inverter-ac.

4. Part replacement — PCB-492 / CAP-492

PCB-492 — Inverter control board assembly (board). Isolate mains and follow the safety SOP. Remove the faulty unit, fit PCB-492, and run the self-test. Requires 'inverter-ac'. This part is covered under active warranty (free). List price ■4,800.

CAP-492 — Compressor capacitor (board). Isolate mains and follow the safety SOP. Remove the faulty unit, fit CAP-492, and run the self-test. Requires 'inverter-ac'. This part is covered under active warranty (free). List price ■3,200.

5. Part replacement — SENSOR-11 / FILTER-100

SENSOR-11 — Outdoor temperature sensor (sensor). Isolate mains and follow the safety SOP. Remove the faulty unit, fit SENSOR-11, and run the self-test. Requires 'general-hvac'. This part is covered under active warranty (free). List price ■1,200.

FILTER-100 — Consumable air filter (consumable). Isolate mains and follow the safety SOP. Remove the faulty unit, fit FILTER-100, and run the self-test. Requires 'general-hvac'. This part is chargeable (out of warranty scope). List price ■450.

6. Part replacement — COMP-492 / FAN-492

COMP-492 — Rotary inverter compressor (compressor). Isolate mains and follow the safety SOP. Remove the faulty unit, fit COMP-492, and run the self-test. Requires 'inverter-ac'. This part is covered under active warranty (free). List price ■18,500.

FAN-492 — Outdoor condenser fan motor (motor). Isolate mains and follow the safety SOP. Remove the faulty unit, fit FAN-492, and run the self-test. Requires 'general-hvac'. This part is chargeable (out of warranty scope). List price ■2,600.

7. Part replacement — VALV-492

VALV-492 — Electronic expansion valve (valve). Isolate mains and follow the safety SOP. Remove the faulty unit, fit VALV-492, and run the self-test. Requires 'inverter-ac'. This part is chargeable (out of warranty scope). List price ■1,900.

8. Maintenance, safety & warranty

Maintenance: inspect at each visit; clean or replace consumables on the service interval; log any recurring fault code.

Discharge the high-voltage capacitor before touching the inverter board. Recover refrigerant; never vent R32.

Warranty annex: under an active warranty, PCB-492, CAP-492, SENSOR-11, COMP-492 are replaced free with labour. Consumables and cosmetic parts are never covered. If warranty is expired or the fault is outside the original scope, the part is chargeable: a quote is raised and payment is collected before the visit is confirmed.

Policy reference: <https://support.example-fieldflow.test/warranty/v5>